

Institute of Actuaries of India

Subject CB1-Business Finance

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1: B [2]

Solution 2: A [2]

Solution 3: C [2]

Solution 4: B [2]

Solution 5: D [2]

Solution 6: B [2]

Solution 7: A [2]

Solution 8: B [2]

Solution 9: B [2]

Solution 10: D [2]

Solution 11:

Assumption : All three methods ignores the time value of money [1]

a)

Time in Years	0	1	2	3
Cashflow	(9)	2	2	6
Initial Investment charge		(3)	(3)	(3)
Annual capital charge		(1)	(1)	3

[1]

Annual capital charge is similar to the accounting depreciation charge calculated using straight line basis method. [1]

The annual capital charge after 3 years is positive, which shows the effect on profitability of the investment after depreciation over a three year time period [1]

[Max 1 mark for the comment]

b) For this project the payback period will be 3 years [1]

It can be considered as positive aspect of this project as the it shows the company need not finance for a long period. [1]

c) Ratio of positive cashflows to negative cashflow over 3-year period, which is $10/9 = 1.11$ [1]

Since it is greater than 1 it can be considered as positive aspect for the project. [1]

[Max 5 Marks]

Solution 12:

Advantages

Shares are likely to remain with a small group of shareholders (those who purchased the company) who retain control of the company. [1]

A less diverse group of owners means that the principal-agent problems are reduced and thus the company can be managed more effectively to meet the objectives of shareholders.

[1]

Fewer rules and regulation mean less onerous disclosure and reporting requirements, in particular, hence cost will be lower. [1]

Disadvantages

When Go-mobile was a quoted company, the company was able to sell shares to the public. Since the company has now become a private company, it will have restricted access to fund (cannot sell shares to the public). [1]

Lenders will not rely on the company (not required to meet the requirements of stock exchange) and borrowing becomes expensive. [1]

Limited market for shares and so a limited exit route for shareholders. This will lead to low liquidity and high transaction cost. [1]

Shares of Go- mobile are not easy to value. [1]

[Max 5 Marks]

Solution 13:

i)

1. Currency futures
2. Currency swaps
3. Interest rate swaps
4. Options

[2]

- ii) ABC is a multinational company, therefore operating in its home country as well as in other countries around the world. The company is likely to have large contracts that involve substantial payments at a future date. The company could use currency futures to “lock in” exchange rates of receipts or payments that have been agreed for a future date. This will protect the company from the risk of liabilities increasing in value or assets falling in value before they fall due to be settled. [3]

The company might use currency swaps to reduce its total cost of borrowing. ABC may have a comparative advantage in borrowing in the main currency in the home country. It may enter into a reciprocal arrangement with a borrower in the host country who has a corresponding comparative arrangement. The two parties can exchange a series of payments so that each has a finance in their desired country and currency, but payable on terms that have been negotiated with their counterparty. [3]

Similarly, the company can use interest rate swaps. ABC can exchange a fixed interest rate for a floating rate with another company which has a corresponding advantage. [2]

ABC can use options to hedge against risk of increase in the prices of raw materials such as steel, aluminium. Thus the company can protect itself from the increase in the input cost. [1]

ABC may provide incentive to its customers (Cruise ship company) by selling them, its own futures, and options. Cruise ship company would take options on cruise ships. Cruise ship company gets the right but not the obligation to purchase at a predetermined date but at fixed price. This means the Cruise ship company is protected from the increase in prices. If the cruise ship company does not buy, ABC can retain the cost of the option. [1]

Explanation- this will help ABC to have more customers as they can buy the cruise at a price already decided.

[Max 8]

[10 Marks]

Solution 14:

- i) Covered bonds are bonds issued by banks or building societies with ring fenced pools of assets that will repay investors if the issuing institutions fall. The ring-fenced assets continue to be legally owned by the issuing banks or building societies unless they fall.

[2]

ii) **Advantages**

1. **Security:** Covered bonds are like debentures in that the company remains legally responsible for the payment of interest and capital to bondholders. [1]
Bond holders have a dual recourse to reclaim their money, firstly from issuing company and then secondly from the pool of assets. [1]
Covered bonds are very secure and often receive a AAA credit rating. [1]
2. **Liquidity:** Since over 1 lakh crores of these bonds have already been issued in the market, there is always an investor who is willing to trade these bonds for cash. This gives Pension fund the ability to enter and exit the market at any time when the need arises. [1]
3. **Regulatory Benefit:** They are the safest asset class in the fixed-income security market So, there might be some regulatory benefits if the regulator incentivizes the holding of these bonds. Lower regulatory requirements if included as an asset class [1]
4. **Since the risk of default is almost negligible,** investors are willing to accept much smaller coupon payment. This is the reason why companies are increasingly issuing covered bonds. Managers will have greater choices of selection of covered bonds suitable to the needs of pension fund. [1]

[Max 4]iii) **Risk**

1. The price of covered bonds depends on many factors. For example, Investors have to analyze the financial condition of the pool of assets backing the bond. In addition, the investor needs be aware of the financial condition of the issuer who is issuing the asset. Credit rating is a lagging indicator of credit quality, investors cannot rely much on credit rating firms. This complexity makes covered bonds difficult to understand. [2]
2. **Low Yields-** As the risk of default is low rather negligible, the issuers do not feel any compulsion to offer higher returns [1]
3. Though the Covered bond are backed by the pool of assets, bond issue is considered as the liability of the company. Hence, if a company issue covered bonds excessively, it may witness a significant fall in its credit quality. This may increase the cost of debt. Therefore, any company can underwrite only a limited number of bonds. [2]
4. During recession period when people are afraid of mass defaults happening, covered bonds have high demand. Therefore, expensive for the pension fund to purchase. Whereas during periods of high economic activity, it is difficult to sell these bonds to raise more money. [2]

[Max 4]**[10 Marks]****Solution 15:**

Capital structure or gearing can affect the profits available to the shareholders. [1]

Debt is relatively cheap source of finance. If the company borrows rather than issuing further share capital then a large proportion of the returns from the resulting investments will be available to the shareholders. [1.5]

The company can also claim tax relief on the interest and that will further decrease the cost of debt. [1]

Debt can also affect the volatility of returns to shareholders. Interest and capital repayment must be paid regardless of whether the company has had a good year or not. The fact that this fixed commitment must be met means that the company may have very little profit left for the shareholders in leaner years. [2]

Conversely, the fixed return offered to the lenders will mean that any upturn in revenue will lead to all of the excess being passed on to the shareholders. This means that borrowing will accentuate any fluctuations in the operating profit. [2]

[Max 6 Marks]

Solution 16:

The current depreciation treatment may increase the company's profits, the auditor may issue a qualified audit report that will state that the financial statements present fairly except for the impact of the depreciation treatment. [1]

- Shareholders may suspect fraud or creative accounting by the company and sell their shares, leading to fall in the share price. [1]
- This could lead to takeover bid by a competitor and may lead to the directors being replaced [1]
- Creditors may be less willing to lend, customers and suppliers being less likely to do business with the company [1]
- Or the Shareholders may agree with the company directors, in which case the share price will not move downwards [1]
- The company may change the external auditor to try to resolve the problem, however this may not be liked by the shareholders. [1]

[Max 4 Marks]

Solution 17:

- i) Shareholders and Managers: The main objective of the shareholders will be to receive high return on their investment in the company. In contrast some managers might instead wish to a) pursue projects of interest over more profitable projects or b) to takeover other companies and form a large business empire c) to have a more leisurely or luxurious working lifestyle [1]
- ii) Shareholders and Providers of finance: Shareholders may be keen to invest in high risk and high return project whereas the providers of finance, lenders such as banks and bondholders have no particular interest in upside profits, they simply want to ensure that they receive the promised interest and capital payments. [1]
- iii) Managers and workers – Managers may be interested in labor saving technology for achieving better work efficiency, but workers fear the loss of job. [1]
- iv) Shareholders and local community – Shareholders may wish to expand production on a particular site but local community and local government may fear greater air pollution and more congestion. [1]
- v) Managers and Customers- Managers may be interested in labour saving technology for achieving better work efficiency, but customers may fear of quality of service. [1]
Manager may knowingly advise customer to purchase products which are not in the best interest of the customers (too expensive too risky) but which help the manager to achieve his monthly target sales. [1]

[Max 5 Marks]

Solution 18:

i)

Operating profit	7,50,000
Depreciation	2,77,000
Increase in inventories	-72,000
Decrease in trade receivables	20,000
Increase in trade payables	11,000

Cash in hand ₹9,86,000

[3]

ii) Balance on revaluation reserve = $58+2-50=10$ lakhs

[2]

[5 Marks]**Solution 19:**

i)

Solution		
	X	Y
Profitability		
Return on capital employed	$579/(2200+920)=19\%$	$1115/(3831+415)=26\%$
Gross profit percent	$(1110/1900)=58\%$	61%
Advertising/revenue	$(310/1900)=16\%$	19%
Liquidity		
Current ratio	$(280/360)=0.8:1$	7.9:1
Efficiency		
Receivables turnover	$(280/1900*365)=54\text{days}$	74 days

Comparing the Return of capital employed ratio both the subsidiaries are strong but Y appears much stronger than X.

The Gross profit is good for both of the subsidiaries and the difference is not significant between the subsidiaries.

Y is getting a higher return on advertising and it may have identified it and so spends higher on advertising compared to X.

The turnover of Y compared to its asset base is higher than that of X. It may indicate that Y is more efficient subsidiary in this angle.

The overdraft of X relative to its current assets appears much higher than Y. If bank calls for immediate repayment of overdraft X will become insolvent.

The Receivables turnover for X is shorter than that of Y which indicates X is efficient in getting payments from the customers promptly. But it may lead to customers finding some new company which allows for longer payment time. This may reduce the new business for X.

The requirement of X to service large overdraft may be a reason why it allows shorter payment period for its customers.

[Max 12]

ii)

The annual report only gives the snapshot as at accounting year end and it may not be representative of the year as a whole but can only give some insight into changes.

For example, the time lag between preparation and publication of the report could result in changes in the company's liquidity position.

Apart from the ratios the director should see whether both the figures are directly comparable.

The assumption used in preparation of the accounting figures may be different between X and Y. For example, Y may be more aggressive in booking the turnover.

The markets in the state where X operates may not be as matured as Y. In this sense X may be performing well and any changes in the operations without understanding the difference in the market may not be useful.

In terms of turnover Y is much higher which implies the economies of scale is in its advantage. Also, Y may be much more stabilized company as it could have longer experience in the market.

The state in which Y operates may have more qualified engineers and hence the availability of suitable staffs would have improved the efficiency.

[Max 4]

[16 Marks]

Solution 20:

- i) Geared beta = ungeared beta $\times$ (1 + debt: equity ratio $\times$ (1 – tax rate))
 $1.25 \times (1 + 260/500 \times (1 - 0.22)) = 1.757$
 Cost of equity = $6 + (1.757 \times (11 - 6)) = 14.785\%$ [4]

- ii) Coupon payment on debt = $320 \times 5\% = 16$
 From the *tables* the values of v^5 and a_5 @ 8% and for 12% are used to calculate NPV
 NPV of debt at 8% = $-260 + (16 \times 3.993) + (320 \times 0.681) = 21.67$
 NPV of debt at 12% = $-260 + (16 \times 3.605) + (320 \times 0.567) = -20.75$
 Yield is approximately half way between 8% and 12% = 10% (rounded to nearest whole %)
 Cost of Debt = $10\% \times (1 - 0.22) = 7.8\%$ [4]

- iii) WACC = $(260/(260 + 500) \times 7.8) + (500/(260 + 500) \times 14.785) = 12.695\%$ [2]

- iv) In general, the debt is cheaper than equity because the lender takes fewer risks.

Theoretically borrowing will reduce WACC because of higher proportion of debt.

In practice the borrowing will affect the gearing levels which in-turn affects the risk characteristics of existing equity.

Borrowing will increase the risk of holding shares and hence that could increase the cost of equity.

Also, at higher levels of gearing the risk attached to debt might also increase which could increase the cost of borrowing.

These additional costs will offset the savings from using the cheaper source of finance.

There is a tax advantage in borrowing because interest can be offset against profit but dividends cannot.

If the taxation is ignored it can be seen that the reduction in WACC due to borrowing will be offset by overall increase in the cost of equity.

[Max 4]

[14 Marks]
